# Appendix A

The PubMed, Web of Science, Scopus and Pro-Quest databases were searched using the following search strings, one at a time:

1. "musculotendinous injury" OR "musculotendinous injuries OR "musculotendinous failure" "musculotendinous tear" OR "muscle-tendon injury" OR "muscle-tendon injuries" OR "musculotendinous defect" OR "musculotendinous junction tear" OR musculotendinous junction injury" OR musculotendinous junction injuries" OR "musculotendinous junction failure" OR "musculotendinous junction defect" OR "muscle-tendon failure" OR "muscle-tendon tear" OR "myotendinous injury" OR "myotendinous injuries" OR "myotendinous tear" OR "myotendinous failure" OR "myotendinous defect" OR "myotendinous junction tear" OR "myotendinous junction failure" OR "myotendinous junction defect" OR "tendon-muscle injuries" OR tendon-muscle injury" OR tendon-muscle failure" OR "tendon-muscle tear" OR "tendon-muscle defect"

2. "tendon-bone injury" OR "tendon-bone injuries" OR "tendon-bone tear" OR "tendon-bone defect" OR "bone-tendon injury" OR "bone-tendon injuries" OR "bone-tendon tear" OR "bone-tendon defect" OR "tendinous avulsion" OR "tendon avulsion" OR "aponeurosis avulsion" OR "aponeurotic avulsion" OR "musculotendinous avulsion" OR "myotendinous avulsion"

3. "muscle belly tear" OR "muscle belly injuries" OR "muscle belly injury" OR "muscle belly failure" OR "muscle belly defect" OR "intramuscular tear" OR "intramuscular failure" OR "intramuscular defect" OR "intramuscular injury" OR "intramuscular injuries" OR "muscle substance tears" OR "muscle substance failure" OR "muscle substance defect" OR "muscle substance injury" OR "muscle substance injuries" OR "muscle tear" OR "muscle defect" OR "muscle failure"

4. "tendon tear" OR "aponeurosis tear" OR "tendinous tear" OR "tendon failure" OR "tendon defect" OR "tendinous failure" OR "tendinous defect" OR "tendinous failure" OR "aponeurosis failure" OR "aponeurosis defect"

# Appendix B

To access quality and risk of bias of the laboratory studies, the ARRIVE[77] guidelines checklist was implemented. Items 9, 14 and 17 were not considered since they were not relevant in the context of all the laboratory studies included in the analysis. To the remaining items, a score was given, depending on the number of details involved. The sum of the points of all items represents the maximum possible score, or a perfect 100% evaluation. Each article was evaluated in these 17 items and a percentual score was determined afterwards. Non-Applicable items were not counted and were deducted from the maximum possible score for that study. The items under evaluation, adapted from Kilkenny, et al 2010[77], and the score for each item it is presented in Table B1 and the evaluation performed and the score for each laboratory study in Table B2.

**Table B1- ARRIVE guidelines checklist and score defined for the evaluation.**

| Item | | Recommendations | Score |
|---|---|---|---|
| **Title** | **1** | Provide an accurate and concise description of the content | 1 |
| **Abstract** | **2** | Provide background, objectives, animal used, methods, results and conclusions | 6 |
| **Introduction** | **3** | Include sufficient scientific background.<br>Explain how and why animal species and model being used can address the scientific objectives. | 2 |
| | **4** | Describe the objectives of the study or hypothesis being tested | 1 |
| **Methods** | **5** | Include ethical review permissions | 1 |
| | **6** | Include the number of experimental and control groups<br>Include steps taken to minimize effects of subjective bias (e.g. randomisation)<br>Define the experimental unit | 3 |
| | **7** | Provide details regarding the anaesthesia formulation and dose, route of administration, surgical procedure and method of euthanasia | 4 |
| | **8** | Provide details of the animals used: species, development stage, weight and source of animals | 4 |
| | **10** | Specify the total number of animals used and how the number of animals was arrived at | 2 |
| | **11** | Give full details of how animals were allocated to experimental groups | 1 |
| | **12** | Define the experimental outcomes assessed | 1 |
| | **13** | Provide details of the statistical methods<br>Specify the unit of analysis for each dataset<br>Describe any methods used to assess whether the data met the assumptions of the statistical approach | 3 |
| **Results** | **15** | Report the number of animals in each group included in the analysis<br>If any animals or data where not included in the analysis explain why | 2 |
| | **16** | Report the results for each analysis carried out with a measure of precision | 1 |
| **Discussion** | **18** | Interpret the results<br>Comment on the study limitations | 2 |
| | **19** | Comment on whether, and how, the findings of this study are likely to translate to other species or systems, including any relevance to human biology | 1 |
| | **20** | List all funding sources | 1 |
| | | **TOTAL** | 36 |

Adapted from Kilkenny, et al 2010[77]

**Table B2 - Quality evaluation and score of the laboratory studies included in this study.**

| | ARRIVE items | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Author** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **10** | **11** | **12** | **13** | **15** | **16** | **18** | **19** | **20** | **TOTAL** | **Max Score** | **% Score** |
| (Garrett et al. 1987)[48] | 1 | 3 | 1 | 1 | 0 | 2 | 3 | 2 | 1 | 0 | 1 | 3 | 2 | 1 | 1 | 0 | 0 | 22 | 35 | 63% |
| (Garrett et al. 1988)[47] | 1 | 2 | 1 | 1 | 0 | 2 | 3 | 2 | 1 | 0 | 0 | 2 | 2 | 1 | 1 | 0 | 0 | 19 | 36 | 53% |
| (Taylor et al. 1993)[143] | 1 | 3 | 1 | 1 | 0 | 2 | 3 | 2 | 1 | 0 | 1 | 2 | 2 | 0 | 1 | 0 | 0 | 20 | 36 | 56% |
| (Noonan et al. 1994)[104] | 1 | 3 | 1 | 1 | 0 | 2 | 2 | 2 | 1 | 0 | 1 | 2 | 0 | 1 | 2 | 0 | 0 | 19 | 36 | 53% |
| (Sun et al. 1994)[139] | 1 | 4 | 1 | 1 | 0 | 3 | 3 | 2 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 0 | 0 | 24 | 36 | 67% |
| (Gottsauner-Wolf et al. 1995)[54] | 1 | 4 | 1 | 1 | 0 | 3 | 0 | 2 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 0 | 1 | 22 | 33 | 67% |
| (Hasselman et al. 1995)[59] | 1 | 5 | 1 | 1 | 0 | 2 | 0 | 2 | 1 | 0 | 1 | 2 | 2 | 1 | 2 | 0 | 1 | 22 | 36 | 61% |
| (Law, Caputo, and Tidball 1995)[79] | 1 | 5 | 1 | 1 | 0 | 1 | 1 | 2 | 1 | 0 | 1 | 2 | 1 | 1 | 1 | 0 | 0 | 19 | 36 | 53% |
| (Mair et al. 1996)[88] | 1 | 3 | 2 | 1 | 0 | 1 | 3 | 1 | 0 | 0 | 1 | 2 | 0 | 1 | 1 | 0 | 0 | 17 | 36 | 47% |
| (Sano et al. 1997)[121] | 1 | 5 | 1 | 1 | 0 | 2 | NA | 2 | 1 | NA | 1 | 2 | 2 | 1 | 1 | 0 | 1 | 21 | 32 | 66% |
| (Tsuang et al. 1998)[149] | 1 | 6 | 2 | 1 | 0 | 1 | 3 | 2 | 0 | 0 | 1 | 2 | 0 | 1 | 1 | 0 | 1 | 22 | 36 | 61% |

NA – non-applicable

For the observational cohort studies, the STROBE[152] guidelines checklist was implemented. Item 17 was not considered since it was not relevant in the included cohort studies. As with the laboratory studies, a score was given to each item, depending on the number of details involved. Once again, Non-Applicable items were not counted and were deducted from the maximum possible score for that study. The items under evaluation, adapted from Vandenbroucke, et al 2007[152], and the score for each item it is presented in Table B3 and the evaluation performed and the score for each laboratory study in Table B4.

**Table B3 - STROBE guidelines checklist and score defined for the evaluation.**

| Item | | Recommendations | Score |
|---|---|---|---|
| **Title and Abstract** | **1** | a) Indicate the study design in the title or abstract<br>b) Provide in the abstract a balanced summary of what was done and what was found | 3 |
| **Introduction** | **2** | Explain the scientific background | 1 |
| | **3** | State specific objectives | 1 |
| **Methods** | **4** | Present the type of study design early in the paper | 1 |
| | **5** | Describe the recruiting site and relevant dates (e.g. periods of recruitment) | 2 |
| | **6** | Give the eligibility criteria and the methods of selection of participants | 2 |
| | **7** | Define what was included and/or excluded in the analysis | 1 |
| | **8** | Give details of methods of assessment | 1 |
| | **9** | Describe any efforts to address potential source of bias | 1 |
| | **10** | Explain how the study size was arrived | 1 |
| | **11** | Explain how quantitative variables were handled in the analysis | 1 |
| | **12** | a) Describe all statistical methods<br>b) Explain how missing data was addressed (e.g. patients lost to follow-up) | 2 |
| **Results** | **13** | a) Report numbers of individuals at each stage of the study (e.g. numbers potentially eligible, confirmed eligible, included in the study, completing follow-up, and analysed)<br>b) Give reasons for non-participation at each stage | 2 |
| | **14** | a) Give characteristics of study participants (sex, age, activity at time of injury)<br>b) Indicate number of participants with missing data<br>c) Summarize follow-up time | 5 |
| | **15** | Report numbers of outcome events or summary measures over time | 1 |
| | **16** | Give unadjusted estimates and, if applicable, confounded-adjusted estimates and their precision | 1 |
| **Discussion** | **18** | Summarize key results | 1 |
| | **19** | Discuss limitations of the study | 1 |
| | **20** | Give a cautious overall interpretation of results | |
| | **21** | Discuss the generalisability of the study results | |
| | **22** | Give the source of funding | 1 |
| | | **TOTAL** | 32 |

Adapted from Vandenbroucke, et al 2007[152]

**Table B4 - Quality evaluation and score of the cohort studies included in this study.**

| | STROBE items | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Author** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | **18** | **19** | **20** | **21** | **22** | **TOTAL** | **Max Score** | **% Score** |
| (Durig et al. 1977)[34] | 2 | 1 | 0 | 0 | 1 | 2 | 0 | 0 | 0 | 0 | 0 | NA | 2 | 4 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 16 | 30 | 53% |
| (Ferretti et al. 1983)[40] | 2 | 1 | 1 | 0 | 0 | 2 | 1 | 1 | 0 | 0 | 1 | NA | 2 | 3 | 1 | NA | 1 | 0 | 1 | 0 | 0 | 17 | 29 | 59% |
| (Ferretti et al. 1985)[41] | 2 | 1 | 0 | 1 | 2 | 2 | 0 | 1 | 0 | 1 | 1 | 2 | 2 | 3 | 1 | NA | 1 | 0 | 1 | 0 | 0 | 21 | 31 | 68% |
| (Nimityongskul, Montague, and Anderson 1988)[102] | 2 | 1 | 0 | 0 | 2 | 1 | 0 | 1 | 0 | 0 | NA | 2 | 2 | 5 | 1 | NA | 1 | 0 | 0 | 0 | 0 | 18 | 30 | 60% |
| (Fleckenstein et al. 1989)[43] | 3 | 1 | 1 | 1 | 0 | 2 | 0 | 1 | 0 | 0 | 1 | 2 | 2 | 2 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 19 | 32 | 59% |
| (Farley et al. 1992)[38] | 3 | 0 | 1 | 0 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 21 | 32 | 66% |
| (Pomeranz and Heidt 1993)[111] | 3 | 1 | 0 | 1 | 1 | 2 | 0 | 1 | 0 | 0 | 1 | 1 | 2 | 3 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 21 | 32 | 66% |
| (Speer, Lohnes, and Garrett 1993)[135] | 2 | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | NA | NA | 2 | 2 | 1 | NA | 1 | 0 | 1 | 0 | 0 | 19 | 28 | 68% |
| (Yoshioka et al. 1994)[165] | 2 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | NA | 1 | 5 | 1 | NA | 1 | 0 | 0 | 0 | 0 | 13 | 29 | 45% |
| (Hughes et al. 1995)[64] | 3 | 1 | 1 | 1 | 2 | 2 | 1 | 0 | 0 | 0 | NA | NA | 2 | 5 | 1 | NA | 1 | 0 | 1 | 0 | 0 | 21 | 28 | 75% |
| (Simonet, Saylor III, and Sim 1995)[130] | 3 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | NA | 2 | 5 | 0 | NA | 1 | 0 | 1 | 0 | 0 | 15 | 29 | 52% |
| (Takami et al. 1995)[142] | 3 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | NA | NA | 2 | 5 | 1 | NA | 1 | 0 | 0 | 0 | 0 | 18 | 28 | 64% |
| (Evans, Bowrey, and Newman 1998)[36] | 3 | 1 | 0 | 0 | 2 | 2 | 0 | 1 | 0 | 0 | NA | NA | 2 | 5 | 1 | NA | 1 | 1 | 1 | 1 | 0 | 21 | 28 | 75% |
| (Maffulli 1998)[87] | 3 | 1 | 0 | 1 | 1 | 2 | 0 | 1 | 0 | 0 | 1 | 2 | 2 | 3 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 23 | 32 | 72% |

**Table B4 - continued**

| | STROBE items | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Author** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | **18** | **19** | **20** | **21** | **22** | **TOTAL** | **Max Score** | **% Score** |
| (Temple et al. 1998)[144] | 3 | 1 | 0 | 1 | 2 | 0 | 0 | 1 | 0 | 0 | NA | NA | 2 | 5 | 1 | NA | 1 | 0 | 0 | 1 | 0 | 18 | 28 | 64% |
| (Connell et al. 1999)[26] | 3 | 1 | 1 | 0 | 2 | 2 | 1 | 1 | 0 | 0 | 1 | NA | 2 | 5 | 1 | NA | 1 | 1 | 1 | 1 | 0 | 24 | 29 | 83% |
| (Carrino et al. 2000)[19] | 3 | 1 | 1 | 1 | 1 | 2 | 0 | 1 | 0 | 0 | NA | NA | 2 | 5 | 1 | NA | 1 | 1 | 1 | 1 | 0 | 22 | 29 | 76% |
| (De Smet and Best 2000)[32] | 3 | 1 | 1 | 0 | 2 | 1 | 0 | 1 | 0 | 0 | NA | NA | 2 | 5 | 1 | NA | 1 | 0 | | 0 | 0 | 18 | 29 | 62% |
| (Karjalainen et al. 2000)[73] | 3 | 1 | 1 | 0 | 0 | 2 | 1 | 1 | 0 | 0 | 1 | NA | 2 | 4 | 1 | NA | 1 | 0 | 1 | 0 | 0 | 19 | 29 | 66% |
| (Panni, Tartarone, and Maffulli 2000)[106] | 3 | 1 | 1 | 1 | 2 | 2 | 0 | 1 | 1 | 0 | 1 | NA | 2 | 5 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 24 | 29 | 83% |
| (Tung et al. 2001)[150] | 3 | 1 | 1 | 0 | 1 | 2 | 0 | 1 | 0 | 0 | NA | NA | 2 | 4 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 19 | 28 | 68% |
| (Weishaupt, Schweitzer, and Morrison 2001)[159] | 3 | 1 | 1 | 1 | 2 | 1 | 0 | 1 | 0 | 0 | NA | NA | 2 | 5 | 1 | NA | 1 | 1 | 1 | 1 | 1 | 23 | 28 | 82% |
| (Williams et al. 2001)[162] | 3 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | NA | NA | 2 | 2 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 16 | 28 | 57% |
| (Bass and Connell 2002)[10] | 3 | 1 | 1 | 1 | 2 | 2 | 0 | 1 | 0 | 0 | NA | NA | 2 | 4 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 21 | 28 | 75% |
| (Bianchi et al. 2002)[14] | 3 | 1 | 1 | 1 | 0 | 2 | 0 | 0 | 0 | 0 | NA | NA | 2 | 5 | 1 | NA | 1 | 0 | 0 | 0 | 0 | 17 | 28 | 61% |
| (Sonnery-Cottet et al. 2002)[134] | 3 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 0 | 0 | 1 | 2 | 2 | 5 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 28 | 32 | 88% |
| (Connell, Jhamb, and James 2003)[25] | 1 | 1 | 1 | 0 | 2 | 2 | 0 | 1 | 0 | 0 | 0 | NA | 2 | 5 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 19 | 29 | 66% |
| (T. K. Kim, Rauh, and McFarland 2003)[78] | 3 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 0 | 1 | 2 | 2 | 4 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 27 | 32 | 84% |

**Table B4 - continued**

| Author | STROBE items | | | | | | | | | | | | | | | | | | | | | TOTAL | Max Score | % Score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 18 | 19 | 20 | 21 | 22 | | | |
| (Mansat, Frankle, and Cofield 2003)[90] | 3 | 1 | 1 | 0 | 1 | 2 | 0 | 1 | 0 | 0 | 0 | NA | 2 | 4 | 1 | NA | 1 | 0 | 0 | 0 | 0 | 17 | 29 | 59% |
| (Äärimaa et al. 2004)[1] | 3 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 0 | 0 | NA | 2 | 2 | 5 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 25 | 31 | 81% |
| (T. M. Cross et al. 2004)[29] | 3 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 0 | 0 | 1 | 2 | 2 | 5 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 27 | 32 | 84% |
| (Freccero and Berkowitz 2006)[45] | 3 | 1 | 0 | 1 | 2 | 2 | 1 | 1 | 0 | 1 | 1 | 2 | 2 | 3 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 24 | 32 | 75% |
| (Ilaslan, Iannotti, and Recht 2007)[66] | 3 | 1 | 0 | 1 | 1 | 2 | 0 | 1 | 1 | 0 | NA | NA | 2 | 3 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 19 | 28 | 68% |
| (Irmola et al. 2007)[67] | 2 | 1 | 0 | 1 | 2 | 2 | 0 | 1 | 0 | 0 | NA | NA | 2 | 4 | 1 | NA | 1 | 0 | 0 | 0 | 0 | 17 | 28 | 61% |
| (Lempainen et al. 2007)[80] | 3 | 1 | 1 | 1 | 2 | 2 | 0 | 1 | 0 | 0 | NA | NA | 2 | 5 | 1 | NA | 1 | 1 | 0 | 0 | 1 | 22 | 28 | 79% |
| (Gyftopoulos et al. 2008)[56] | 2 | 1 | 1 | 1 | 2 | 2 | 0 | 1 | 0 | 0 | 1 | NA | 2 | 3 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 20 | 29 | 69% |
| (Lunn et al. 2008)[85] | 3 | 1 | 1 | 1 | 2 | 2 | 0 | 0 | 0 | 0 | 0 | 2 | 2 | 5 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 24 | 32 | 75% |
| (Silder et al. 2008)[129] | 3 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 2 | 2 | 5 | 1 | NA | 1 | 0 | 0 | 0 | 1 | 21 | 31 | 68% |
| (Gaskin et al. 2009)[49] | 3 | 1 | 1 | 1 | 0 | 2 | 1 | 1 | 0 | 0 | NA | NA | 2 | 4 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 20 | 28 | 71% |
| (Basic-Jukic et al. 2009)[9] | 3 | 1 | 0 | 0 | 2 | 2 | 0 | 1 | 0 | 0 | 1 | NA | 2 | 4 | 1 | NA | 1 | 0 | 0 | 0 | 0 | 18 | 29 | 62% |
| (Walch et al. 2009)[156] | 3 | 1 | 1 | 0 | 1 | 2 | 0 | 1 | 0 | 0 | 0 | 2 | 2 | 4 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 19 | 32 | 59% |
| (Pochini et al. 2010)[110] | 3 | 1 | 0 | 1 | 1 | 2 | 1 | 1 | 1 | 0 | 1 | 2 | 2 | 5 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 26 | 32 | 81% |

**Table B4 - continued**

| | STROBE items | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Author** | **1** | **2** | **3** | **4** | **5** | **6** | **7** | **8** | **9** | **10** | **11** | **12** | **13** | **14** | **15** | **16** | **18** | **19** | **20** | **21** | **22** | **TOTAL** | **Max Score** | **% Score** |
| (Pedret et al. 2011)[107] | 2 | 1 | 0 | 1 | 1 | 2 | 0 | 1 | 0 | 0 | 1 | NA | 2 | 5 | 1 | NA | 1 | 0 | 0 | 0 | 0 | 18 | 29 | 62% |
| (Makridis et al. 2014)[89] | 3 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 0 | 0 | 0 | 2 | 1 | 4 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 23 | 32 | 72% |
| (Crema et al. 2016)[28] | 3 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 1 | 0 | 0 | 2 | 2 | 5 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 28 | 32 | 88% |
| (Lukac et al. 2016)[84] | 3 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 0 | 0 | 0 | NA | 2 | 4 | 1 | NA | 1 | 1 | 0 | 0 | 0 | 21 | 29 | 72% |
| (Entwisle et al. 2017)[35] | 3 | 1 | 1 | 0 | 1 | 2 | 0 | 1 | 0 | 0 | 1 | NA | 2 | 5 | 1 | NA | 1 | 1 | 1 | 0 | 0 | 21 | 29 | 72% |
| (Godoy et al. 2018)[52] | 3 | 1 | 1 | 1 | 0 | 2 | 1 | 1 | 0 | 0 | 1 | 2 | 2 | 5 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 25 | 32 | 78% |
| (Roe et al. 2018)[117] | 3 | 1 | 1 | 0 | 2 | 1 | 0 | 1 | 0 | 0 | 0 | 2 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 15 | 32 | 47% |
| (Miranda and Bureau 2019)[97] | 3 | 1 | 1 | 1 | 2 | 1 | 1 | 1 | 0 | 0 | 1 | 2 | 2 | 3 | 1 | NA | 1 | 1 | 1 | 1 | 0 | 24 | 31 | 77% |

NA – non-applicable

Finally, for the case report studies, the CARE[46] guidelines checklist was implemented. Items 2, 7 and 12 were not considered. The system of points for each item was defined as with the previous checklists. The items under evaluation, adapted from Gagnier et al 2013[46], and the score for each item it is presented in Table B5 and the evaluation performed and the score for each laboratory study in Table B6.

**Table B5 - CARE guidelines checklist and score defined for the evaluation.**

| Item | | Recommendations | Score |
|---|---|---|---|
| **Title** | **1** | The words Case Report should be in the title along with the area of focus | 1 |
| **Abstract** | **3** | Describe:<br>a) what is unique about the case<br>b) the main symptoms of the patient<br>c) the main diagnosis, therapeutic interventions, and outcomes<br>d) what are the main "take-away" lessons from this case | 4 |
| **Introduction** | **4** | One or two paragraphs summarizing why this case is unique with references | 1 |
| **Patient Information** | **5** | a) De-identified demographic information and other patient specific information<br>b) Main concerns and symptoms of the patient<br>c) Medical, family, and psychosocial history including relevant genetic information<br>d) Relevant past intervention and their outcomes | 4 |
| **Clinical Findings** | **6** | Describe the relevant physical examinations and other significant clinical findings | 1 |
| **Diagnostic Assessment** | **8** | a) Diagnostic methods<br>b) Diagnostic reasoning | 2 |
| **Therapeutic Intervention** | **9** | a) Types of intervention (e.g. pharmacological, surgical, conservative,…)<br>b) Administration of intervention | 2 |
| **Follow-up and Outcomes** | **10** | a) Clinician and patient-assessed outcomes<br>b) Important follow-up diagnostic and other results<br>c) Intervention adherence and tolerability<br>d) Adverse and unanticipated events | 4 |
| **Discussion** | **11** | a) Discussion of the strengths and limitations in your approach to this case<br>b) Discussion of the relevant medical literature<br>c) The rationale for conclusions<br>d) The primary take-away lessons of this case report | 4 |
| **Informed consent** | **13** | Indicate if the patient provided informed consent | 1 |
| | | **TOTAL** | 24 |

Adapted from Gagnier et al 2013[46].

**Table B6 - Quality evaluation and score of the case report studies included in this study.**

| | CARE items | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Author** | **1** | **3** | **4** | **5** | **6** | **8** | **9** | **10** | **11** | **13** | **TOTAL** | **Max Score** | **% Score** |
| (Wenger 1973)[160] | 0 | 4 | 0 | 2 | 0 | 2 | 1 | 2 | 2 | 0 | 13 | 24 | 54% |
| (Kawashima et al. 1975)[74] | 1 | 1 | 0 | 2 | 1 | 1 | 2 | 4 | 3 | 0 | 15 | 24 | 63% |
| (Shields and Ashby 1975)[127] | 0 | 0 | 1 | 2 | 1 | 1 | 1 | 4 | 3 | 0 | 13 | 24 | 54% |
| (Miller 1977)[96] | 0 | 0 | 1 | 3 | 1 | 0 | 2 | 3 | 4 | 0 | 14 | 24 | 58% |
| (Bowers 1981)[15] | 0 | 2 | 1 | 2 | 1 | 1 | 2 | 2 | 3 | 0 | 14 | 24 | 58% |
| (Farrar et al. 1981)[39] | 0 | 0 | 1 | 4 | 1 | 2 | 2 | 3 | 4 | 0 | 17 | 24 | 71% |
| (Smith 1981)[132] | 1 | 0 | 1 | 1 | 1 | 2 | 1 | 0 | 1 | 0 | 8 | 24 | 33% |
| (Vastamäki, Brummer, and Solonen 1981)[154] | 1 | 1 | 1 | 1 | 0 | 2 | 1 | 3 | 3 | 0 | 13 | 24 | 54% |
| (Aso and Torisu 1984)[6] | 0 | 2 | 1 | 2 | 1 | 2 | 1 | 1 | 3 | 0 | 13 | 24 | 54% |
| (Sherman, Snyder, and Fox 1984)[126] | 1 | 0 | 1 | 4 | 1 | 2 | 1 | 2 | 4 | 0 | 16 | 24 | 67% |
| (White and Riley Jr. 1985)[161] | 1 | 0 | 1 | 3 | 1 | 2 | 1 | 2 | 4 | 0 | 15 | 24 | 63% |
| (Andersen 1986)[3] | 0 | 0 | 0 | 1 | 1 | 1 | 2 | 2 | 4 | 0 | 11 | 24 | 46% |
| (Bach, Warren, and Wickiewicz 1987))[7] | 1 | 0 | 1 | 3 | 1 | 1 | 2 | 4 | 4 | 0 | 17 | 24 | 71% |
| (de Waal Malefijt and Beeker 1987)[33] | 1 | 3 | 1 | 3 | 1 | 2 | 1 | 2 | 2 | 0 | 16 | 24 | 67% |
| (Clay 1988)[24] | 0 | 0 | 0 | 2 | 1 | 2 | 1 | 2 | 4 | 0 | 12 | 24 | 50% |
| (Menz and Nettle 1989)[93] | 0 | 0 | 1 | 3 | 1 | 1 | 2 | 2 | 4 | 0 | 14 | 24 | 58% |
| (Gruel 1990)[55] | 0 | 2 | 1 | 2 | 1 | 2 | 1 | 0.5 | 4 | 0 | 13.5 | 24 | 56% |
| (de Roos and Zeeman 1991)[31] | 0 | 2 | 1 | 2 | 1 | 1 | 2 | 1 | 2 | 0 | 12 | 24 | 50% |
| Tiger, Mayer, and Glazer 1993)[146] | 0 | 3 | 1 | 2 | 1 | 2 | 1 | 0 | 3 | 0 | 13 | 24 | 54% |
| (Goodier, Maffulli, and Good 1994)[53] | 0 | 1 | 1 | 3 | 1 | 2 | 2 | 2 | 4 | 0 | 16 | 24 | 67% |
| (Butcher, Siekanowicz, and Pettrone 1996)[17] | 0 | 2 | 1 | 3 | 1 | 2 | 1 | 2 | 4 | 0 | 16 | 24 | 67% |
| (Ohashi et al. 1996)[105] | 0 | 3 | 1 | 2 | 1 | 2 | 1 | 0 | 4 | 0 | 14 | 24 | 58% |
| (Wagner and Cooney 1997)[155] | 1 | 2 | 1 | 3 | 1 | 2 | 2 | 1 | 3 | 0 | 16 | 24 | 67% |
| (Asinger and el-Khoury 1998)[5] | 0 | 0 | 1 | 4 | 1 | 2 | 1 | 1 | 4 | 0 | 14 | 24 | 58% |
| (Canovas, Nicolau, and Bonnel 1998)[18] | 0 | 2 | 1 | 2 | 1 | 2 | 1 | 1 | 4 | 0 | 14 | 24 | 58% |

**Table B6 - continued**

| Author | CARE items | | | | | | | | | | TOTAL | Max Score | % Score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 3 | 4 | 5 | 6 | 8 | 9 | 10 | 11 | 13 | | | |
| (Hertel and Lambert 1998) | 0 | 0 | 1 | 4 | 1 | 2 | 2 | 4 | 4 | 0 | 18 | 24 | 75% |
| (Sollender, Rayan, and Barden 1998)[133] | 0 | 3 | 1 | 3 | 1 | 2 | 2 | 3 | 3 | 0 | 18 | 24 | 75% |
| (Takami et al. 1998)[141] | 0 | 2 | 1 | 3 | 1 | 1 | 1 | 1 | 2 | 0 | 12 | 24 | 50% |
| (Cooper and Selesnick 2000)[27] | 1 | 0 | 1 | 3 | 1 | 2 | 2 | 3 | 4 | 0 | 17 | 24 | 71% |
| (Henry and Scerpella 2000)[62] | 1 | 0 | 1 | 3 | 1 | 2 | 2 | 3 | 4 | 0 | 17 | 24 | 71% |
| (Kameyama and Shiraishi 2000)[71] | 1 | 0 | 1 | 3 | 1 | 2 | 2 | 4 | 4 | 0 | 18 | 24 | 75% |
| (Street and Burks 2000)[137] | 1 | 0 | 1 | 4 | 1 | 3 | 1 | 12 | 4 | 0 | 27 | 24 | 113% |
| (Rose and Frassica 2001)[119] | 1 | 0 | 1 | 4 | 1 | 2 | 2 | 4 | 4 | 0 | 19 | 24 | 79% |
| (Pina, Garcia, and Sabater 2002)[108] | 0 | 4 | 1 | 2 | 1 | 2 | 2 | 3 | 4 | 0 | 19 | 24 | 79% |
| (Rajasekhar, Kakarlapudi, and Bhamra 2002)[113] | 0 | 0 | 1 | 2 | 1 | 2 | 2 | 1 | 4 | 0 | 13 | 24 | 54% |
| (Singh and Pooley 2002)[131] | 1 | 2 | 1 | 3 | 1 | 2 | 2 | 1 | 4 | 0 | 17 | 24 | 71% |
| (Toczylowski et al. 2002)[147] | 1 | 0 | 1 | 3 | 1 | 1 | 2 | 3 | 3 | 0 | 15 | 24 | 63% |
| (Michael James DePalma MD 2004)[30] | 0 | 3 | 1 | 3 | 1 | 2 | 1 | 3 | 3 | 0 | 17 | 24 | 71% |
| (Maehara and Sakaguchi 2004)[86] | 1 | 0 | 1 | 1 | 1 | 2 | 2 | 4 | 3 | 0 | 15 | 24 | 63% |
| (Mellado et al. 2004)[92] | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 4 | 4 | 0 | 17 | 24 | 71% |
| (Warme and Whitaker 2004)[158] | 1 | 2 | 1 | 4 | 1 | 2 | 2 | 1 | 4 | 1 | 19 | 24 | 79% |
| (Giuffre and Lisle 2005)[51] | 0 | 2 | 0 | 2 | 1 | 2 | 1 | 0 | 4 | 0 | 12 | 24 | 50% |
| (Wang et al. 2005)[157] | 1 | 3 | 0 | 2 | 1 | 2 | 1 | 2 | 4 | 0 | 16 | 24 | 67% |
| (Kankaya et al. 2006)[72] | 0 | 1 | 0 | 2 | 1 | 2 | 2 | 1 | 1 | 0 | 10 | 24 | 42% |
| (Mechrefe, Walsh, and DiGiovanni 2006)[91] | 1 | 0 | 1 | 2 | 1 | 2 | 2 | 3 | 4 | 0 | 16 | 24 | 67% |
| (Rochman, SanGiovanni, and Uribe 2006)[116] | 1 | 0 | 0 | 2 | 1 | 2 | 1 | 1 | 2 | 0 | 10 | 24 | 42% |
| (Sierra et al. 2006)[128] | 1 | 0 | 1 | 3 | 1 | 2 | 2 | 4 | 3 | 0 | 17 | 24 | 71% |
| (Gidwani and Bircher 2007)[50] | 1 | 3 | 1 | 2 | 1 | 1 | 2 | 1 | 4 | 1 | 17 | 24 | 71% |
| (Kibuule and Fehringer 2007)[76] | 1 | 0 | 1 | 2 | 1 | 2 | 2 | 1 | 4 | 0 | 14 | 24 | 58% |
| (Nishida et al. 2007)[103] | 1 | 3 | 1 | 2 | 1 | 2 | 1 | 3 | 4 | 0 | 18 | 24 | 75% |

**Table B6 - continued**

| | CARE items | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Author** | **1** | **3** | **4** | **5** | **6** | **8** | **9** | **10** | **11** | **13** | **TOTAL** | **Max Score** | **% Score** |
| (Levine and Savoie III 2008)[81] | 0 | 0 | 1 | 4 | 1 | 2 | 2 | 1 | 4 | 0 | 15 | 24 | 63% |
| (Schache et al. 2008)[122] | 1 | 4 | 1 | 2 | 1 | 2 | 2 | 2 | 4 | 0 | 19 | 24 | 79% |
| (Winblad, Escobedo, and Hunter 2008)[163] | 0 | 0 | 1 | 2 | 1 | 2 | 0 | 0 | 2 | 0 | 8 | 24 | 33% |
| (Shah, Montgomery, and Bastidas 2009)[123] | 0 | 0 | 1 | 2 | 1 | 2 | 2 | 2 | 4 | 0 | 14 | 24 | 58% |
| (Floor, Van Der Veen, and Devilee 2010)[44] | 0 | 4 | 1 | 4 | 1 | 2 | 2 | 2 | 4 | 0 | 20 | 24 | 83% |
| (Hasegawa and Schofer 2010)[58] | 1 | 4 | 1 | 2 | 1 | 2 | 1 | 1 | 4 | 0 | 17 | 24 | 71% |
| (Pitts, Garner, and Ortiguera 2010)[109] | 1 | 0 | 1 | 2 | 1 | 2 | 1 | 3 | 4 | 0 | 15 | 24 | 63% |
| (Jansen, Cormier, and Patel 2012)[69] | 1 | 0 | 1 | 2 | 1 | 2 | 2 | 3 | 4 | 1 | 17 | 24 | 71% |
| (Khiami et al. 2012)[75] | 0 | 4 | 1 | 2 | 1 | 2 | 2 | 3 | 4 | 0 | 19 | 24 | 79% |
| (Yoon et al. 2012)[164] | 1 | 0 | 1 | 2 | 1 | 2 | 2 | 3 | 4 | 0 | 16 | 24 | 67% |
| (Hart 2013)[57] | 0 | 0 | 0 | 2 | 1 | 1 | 2 | 2 | 0 | 0 | 8 | 24 | 33% |
| (Naito et al. 2013)[99] | 1 | 3 | 1 | 1 | 1 | 2 | 1 | 2 | 3 | 0 | 15 | 24 | 63% |
| (Valente, Mancuso, and Alecci 2013)[151] | 0 | 3 | 1 | 2 | 1 | 2 | 2 | 2 | 4 | 0 | 17 | 24 | 71% |
| (Benazzo, Marullo, and Pietrobono 2014)[11] | 0 | 1 | 1 | 3 | 1 | 2 | 2 | 3 | 1 | 1 | 15 | 24 | 63% |
| (Huri et al. 2014)[65] | 1 | 4 | 1 | 2 | 1 | 2 | 2 | 2 | 4 | 1 | 20 | 24 | 83% |
| (Sharma et al. 2014)[124] | 0 | 3 | 1 | 2 | 1 | 2 | 2 | 2 | 4 | 0 | 17 | 24 | 71% |
| (Bunshah et al. 2015)[16] | 0 | 4 | 1 | 3 | 1 | 2 | 2 | 3 | 4 | 1 | 21 | 24 | 88% |
| (Cetinkaya et al. 2015)[21] | 0 | 4 | 1 | 2 | 1 | 2 | 2 | 1 | 4 | 1 | 18 | 24 | 75% |
| (Fitzpatrick, Cagle, and Flatow 2016)[42] | 1 | 2 | 0 | 2 | 1 | 1 | 2 | 2 | 4 | 0 | 15 | 24 | 63% |
| (Jaiswal et al. 2016)[68] | 1 | 2 | 1 | 2 | 1 | 1 | 2 | 2 | 4 | 0 | 16 | 24 | 67% |

# Appendix C

**Table C1- Characteristics of the included laboratory studies.**

| Author | no.of specimens | Model | Study Design | Muscle/Tendon (proportion) | Injury Site (proportion) | Major Findings |
|---|---|---|---|---|---|---|
| (Garrett et al. 1987)[48] | 48 | New Zealand White Rabbit | EDL stretched to failure: passive and stimulated at 64 Hz, at 16 Hz | EDL (1.00) | MTJ (1.00) | > UTF in stimulated muscles; No difference in UTD ; All EDL tear at MTJ |
| (Garrett et al. 1988)[47] | 198 | New Zealand White Rabbit | Five architecturally different muscles stretched to failure at different strain rates | EDL (0.18) TA(0.18) P(0.40) RF(0.14) GST(0.09) | MTJ (1.00) | All muscles fail at MTJ; rate of stretch doesn't influence site of injury; UTS > muscles with more pennate structure |
| (Taylor et al. 1993)[143] | 24 | New Zealand White Rabbit | Control EDLs stretched to failure. Tested EDLs subjected to nondisruptive stretch (prior to failure of the control) followed by stretch until failure | EDL (1.00) | MTJ (1.00) | Nondisruptive injury lowered 37% the UTF and 21% the UTD compared with controls. Histology showed incomplete disruptions at MTJ. |
| (Noonan et al. 1994)[104] | 56 | New Zealand White Rabbit | Control EDLs and TAs stretched to failure. Passive stretch of EDLs and TAs to 20% and 30% of the control UTF followed by stretch until failure. | EDL (0.50) TA (0.50) | MTJ (1.00) | Histology of injury at 30% UTF show areas of fiber rupture and hemorrhage near the MTJ. Failure parameters not influenced by neither 20% or 30% UTF injury. |
| (Sun et al. 1994)[139] | 48 | New Zealand White Rabbit | GST muscles stretched to failure at different strain rates. | TS (1.00) | M, within 0.1-30mm of the MTJ (1.0) | Muscle-tendon units show viscoelastic behaviour. All failures occur at or near the MTJ independently of strain rate. > UTF with faster strain rates. |
| (Gottsauner-Wolf et al. 1995)[54] | 20 | Adult Mongrel Dogs | SP were stretched to failure at 10 cm/sec | SP (1.00) | MTJ (1.00) | All SP fail at MTJ; Histology show loss of intracellular staining capacity to Masson's stain, tears and cracks disrupting the intracellular structures, and loss of striation. |
| (Hasselman et al. 1995)[59] | 204 | New Zealand White rabbits | Control EDLs and TAs actively stretched to failure. Active stretch of EDLs and TAs to 60%, 70%, 80% and 90% of the UTF of the ipsilateral control | EDL (0.50) TA (0.50) | MTJ (1.00) | Histology in the 70% group showed focal hemorrhage at the distal MTJ in both muscles. In the 80% and 90%, hemorrhage occurred at the distal MTJ and in muscle fibers near the MTJ. 60% histology showed no damage, 70% showed hemorrhage, acute inflammatory cells, edema, and focal muscle fiber disruption. 80% and the 90% groups showed hemorrhage and a number of lightly stained fibers and more spread damage. |
| (Law, Caputo, and Tidball 1995)[79] | 14 | Mouse | EDLs stretched passively and actively to failure at 2cm/s | EDL (1.00) | M, near MTJ (1.00) | No significant difference in UTS and UTSS between active and passive EDLs. Muscle fibers failed transversely |

**Table C1- continued**

| Author | no.of specimens | Model | Study Design | Muscle/Tendon (proportion) | Injury Site (proportion) | Major Findings |
|---|---|---|---|---|---|---|
| (Mair et al. 1996)[88] | 96 | New Zealand White Rabbit | Control EDLs stretched to failure. EDLs were fatigued by setting the muscle to the length of maximal isometric tetanic force until force dropped 25% or 50%, followed by stretch until failure. | EDL (1.00) | NR | Fatigued muscles absorb less 8-31% energy than control muscles. UTD did not change between control and fatigued EDLs. |
| (Sano et al. 1997)[121] | 31 | Cadaveric Human (mean age 62 yo) | SPT was divided in 3 strips and the middle strip was loaded to failure at 25.4 mm/s. | SPT (1.00) | TB (0.65) T (0.35) | Degenerative changes at the SPT insertion reduce its tensile strength. Histology on tested specimens showed that thin fibers on the articular side of the supraspinatus tendon seemed to contribute to the disruption of fibers. 10 of 20 strips showed avulsion of tendon fibers at the sites of granulation tissue |
| (Tsuang et al. 1998)[149] | 60 | New Zealand White Rabbit | TS of one limb was cyclically loaded for one hour at a rate of 0.5cm/min to 12, 20 and 25% strain. After, it was loaded at 0.5cm/min until failure. The TS of the other limb was stretched to failure at 0.5cm/min. | TS (1.00) | 0.1-1mm of MTJ in SL, 5-10mm of MTJ in GST lateral head, 15-30mm of MTJ in GST medial head | UTD is affected by cyclic stretches at 20%. Load-deformation curve significantly change for cyclic stretches of 25%. After 60 min cyclic stretch at 25% strain, UTF decreased 10%, UTD decreased 20% and the slope of load-deformation increased 28%. Injury can be reproduced at 25% strain of the TS muscle tendon unit. |

M: muscle belly; MTJ: musculotendinous junction ; T: tendon ; TB: tendon-bone ; UTF: ultimate tensile force; UTD: ultimate tensile displacement; UTS: ultimate tensile strain ; UTSS: ultimate tensile stress

EDL: extensor digitorum longus ; P: peroneus; RF: rectus femoris; GST: gastrocnemius; TA: tibialis anterior; SP: supraspinatus ; SPT: supraspinatus tendon ; TS: triceps surae ; SL: soleus ; ACT: achilles tendon

# Appendix D

**Table D1 - Characteristics of the included observational studies.**

| Author | No. of patients | Age | No. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Wenger 1973)[160] | 4 | 17.8 ± 0.5 | 4 (1.00) | NR | PT (1.00) | AV (1.00) | X-ray (0.5), surgery (1.00) | American football (1.00) | inability to flex distal interphalangeal joint |
| (Kawashima et al. 1975)[74] | 2 | 25.5 ± 3.5 | 2 (1.00) | NR | PM (1.00) LD (0.5) | M (0.5) MTJ (1.0) | X-ray (0.5), surgery (1.00) | work accident (1.00) | both M and MTJ injuries showed tenderness, palpable gap and hematoma |
| (Shields and Ashby 1975)[127] | 2 | 20.5 ± 12.0 | 2 (1.00) | NR | PAT (1.00) | AV (1.00) | X-ray (1.00) and surgery (1.00) | high jump (1.00) | swelling of the knee |
| (Miller 1977)[96] | 6 | 45.3 ± 9.7 | 2 (0.33) | NR | GST (1.00) | MTJ (1.00) | physical examination, surgery (1.00) | tennis: sudden dorsiflexion of the ankle with knee extended (1.00) | Defect in the medial head of the GST, palpable and tender, and small mass |
| (Durig et al. 1977)[34] | 19 | 49.0 (33-64) | 12 (0.63) | NR | GST (1.00) | MTJ (1.00) | physical examination, surgery (1.00) | maximal dorsiflexion of the foot with full extension of the knee (1.00) | Palpable dehiscence in the MTJ; proximal displacement of the medial head of the GST; decreased muscle tone; hematoma and edema |
| (Bowers 1981)[15] | 1 | 13.0 | 1 (1.00) | Osgood-Schlatter's disease (1.00) | PAT (1.00) | AV (1.00) | X-ray (1.00), surgery (1.00) | basketball (1.00) | inability to extend the knee, swelling |
| (Farrar et al. 1981)[39] | 3 | 48 ± 16.2 | 2 (0.67) | chronic renal failure (0.33) | TT (1.00) | AV (1.00) | X-ray (1.00), surgery (1.00) | car accident (0.33) fall (0.67) | swelling; palpable depression proximal to the olecranon; flecks of avulsed osseous material. |
| (Smith 1981)[132] | 1 | 36.0 | 1 (1.00) | NR | PT (1.00) | AV (1.00) | X-ray (1.00), surgery (1.00) | fall (1.00) | swelling, inability to flex distal joint |
| (Vastamäki, Brummer, and Solonen 1981)[154] | 5 | 47 ± 10.7 | 5 (1.00) | NR | BT (1.00) | AV (1.00) | surgery (1.00) | skiing (0.2), lift heavy load (0.4), teaching gymnastic (0.2), exercise on horizontal bar (0.2) | diminish forearm flexion or supination, hematoma |
| (Ferretti et al. 1983)[40] | 15 | 22.1 ± 2.4 | 15 (1.00) | anabolic steroid injections (0.33) | PAT (0.78) QT (0.22) | TB (1.00) | NR | volleyball (0.6), basketball (0.2), high jump (0.07), soccer (0.07), ballet (0.07) | tenderness over patellar insertion of the PAT or the QT |

**Table D1 - continued**

| Author | No. of patients | Age | No. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Aso and Torisu 1984)[6] | 2 | 32 ± 5.7 | 1 (0.5) | NR | TR (1.00) | M (1.00) | X-ray (1.00), surgery (0.50), CT (0.5) | kendo (0.5), volleyball (0.5) | palpable soft mass in the posterior part of the arm, tenderness, hematoma |
| (Sherman, Snyder, and Fox 1984)[126] | 1 | 34.0 | 1 (1.00) | professional body builder | TT (1.00) | AV (1.00) | X-ray (1.00), surgery (1.00) | fall: strike on the wall (1.00) | swelling, retraction of the TT insertion |
| (Ferretti et al. 1985)[41] | 110 | NR | 15 (0.14) | steroid injections (NR). All patients are athletes | PAT (0.65) QT (0.25) TBT (0.10) | TB (1.00) | X-ray, physical examination (1.00) | sports activity (1.00). Specifics NR | local tenderness, localized puffiness |
| (White and Riley Jr. 1985)[161] | 1 | 12.0 | 1 (1.00) | NR | SCT (1.00) | AV (1.00) | X-ray (1.00) | climbing a wall (1.00) | avulsion of bone fragment, tenderness at the anterior part of the glenohumeral region |
| (Andersen 1986)[3] | 1 | 44.0 | 1 (1.00) | NR | TT (1.00) | AV (1.00) | X-ray (1.00), surgery (1.00) | parachute landing on outstretched arm (1.00) | tenderness around the elbow, fragments of bone at medial epicondyle of the humerus and small fragment proximal to the olecranon |
| (Bach, Warren, and Wickiewicz 1987)[7] | 1 | 33.0 | 1 (1.00) | anabolic steroid injections (1.00) | TR (1.00) | MTJ (1.00) | physical examination (1.00), surgery (1.00) | weightlifting (1.00) | palpable defect and scar formation at MTJ |
| (de Waal Malefijt and Beeker 1987)[33] | 1 | 66.0 | 1 (1.00) | renal insufficiency treated with hemodialysis | TT (1.00) | AV (1.00) | physical examination (1.00), surgery (1.00) | NR | ecchymosis and subcutaneous soft tissue defect, degeneration of the distal end of the tendon, calcification |
| (Clay 1988)[24] | 1 | 24.0 | 1 (1.00) | NR | ED (1.00) EIT (1.00) | MTJ (1.00) AV (1.00) | surgery (1.00) | work accident (1.00) | inability to extend index finger, tender lump, mass on the back of the hand |
| (Nimityongskul, Montague, and Anderson 1988)[102] | 8 | 15.7 (14.2-16.4) | 7 (0.88) | Osgood-Schlatter's disease (0.75) | TBT (1.00) | AV (1.00) | NR | NR | NR |
| (Fleckenstein et al. 1989)[43] | 19 | NR | 0 (0.00) | N | lower extremities (specific muscle and proportion NR) | MTJ (0.52) | Positive MRI (1.00) | running a 30k race (0.52) | Signs of hemorrhage and edema |
| (Menz and Nettle 1989)[93] | 1 | 16.0 | 1 (1.00) | fracture of the distal tibial shaft 4 years prior to MTJ injury (1.00) | EHL (1.00) | MTJ (1.00) | physical examination (1.00), surgery (1.00) | football (1.00) | inability to extend the toe, ischaemic changes in the muscle and central fibrous tissue |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Gruel 1990)[55] | 2 | 15.5 ± 0.7 | 2 (1.00) | NR | POPT (1.00) | AV (1.00) | physical examination, X-ray, surgery (1.00) | skiing (0.5) American football (0.5) | small bone fragment in the region of the lateral condyle, tense or large effusion, hemarthrosis |
| (de Roos and Zeeman 1991)[31] | 1 | 20.0 | 1 (1.00) | N | FPT (1.00) | T (1.00) | physical examination, surgery (1.00) | fight (1.00) | swelling at the palm of the hand, absence of function of the small finger |
| (Farley et al. 1992)[38] | 100 | 56.0 | 100 (1.00) | N | NR (0.70), SPT (0.18), SPT, TMT and IPT (0.09), SCT (0.04) | T (0.30) | MRI (1.00), surgery (1.00) | NR | muscular atrophy, retraction of the MTJ, subacromial fluid, tendinous thinning, for full thickness tears of SPT, SPT and TMT/IPT and SCT (0.3). Tendinitis, tendon degeneration, partial thickness tear of RC (0.7) |
| (Pomeranz and Heidt 1993)[111] | 14 | (20-29) | NR | professional athletes | QF (0.14) BF (0.43) SM (0.36) ST (0.07) | MTJ (0.43) M (0.5) T (0.07) | MRI (1.00) | professional athletes (1.00) specific sport NR | M injuries showed edema and hemorrhage, MTJ showed fluid collections. |
| (Speer, Lohnes, and Garrett 1993)[135] | 50 | 28.0 (17-42) | NR | N | ADL (0.26) BF (0.22) RF (0.20) GST (0.20) SM (0.08) ST (0.04) | MTJ (1.00) | Positive MRI (0.46) Positive CT scan (0.54) | NR | Consistently seen inflammation and edema. CT showed signs of hemorrhage and proximal retraction |
| Tiger, Mayer, and Glazer 1993)[146] | 1 | 52.0 | 1 (1.00) | NR | TT (1.00) | AV (1.00) | MRI (1.00) | fall (1.00) | tenderness, edema, tendon retraction, hemorrhage |
| (Goodier, Maffulli, and Good 1994)[53] | 1 | 18.0 | 1 (1.00) | N | PAT (1.00) | AV (1.00) | X-ray (1.00), surgery (1.00) | soccer (1.00) | hemarthrosis, avulsion of bone fragment, palpable defect at TBT site |
| (Yoshioka et al. 1994)[165] | 7 | 22.7 ± 4.8 | NR | well trained athletes | BF (0.71) RF (0.14) RF and VM (0.14) | MTJ (0.86) M (0.29) | Positive MRI (1.00) | sprinting (0.43) high jump (0.14) long jump (0.14) tennis (0.14) basketball (0.14) | Fan like lesion in MTJ and signs of hemorrhage and tissue edema |
| (Hughes et al. 1995)[64] | 10 | 22.7 ± 7.6 | 2 (0.20) | N | RF (1.00) | MTJ (1.0) | Positive MRI (0.8) Positive CT scan (0.3) | sprinting (0.70) placekicking (0.10) NR (0.20) | Palpable mass of soft tissue in RF; The mass appears as a bull's eye with a round enhancing mass centered about the intramuscular tendon of the indirect head. Proximal retraction of the torn muscle fibers |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Simonet, Saylor III, and Sim 1995)[130] | 10 | 25.5 (28-32) | 10 (1.00) | professional hockey players | IAB (1.00) | MTJ (1.00) | physical examination (1.00), surgery (1.00) | ice hockey (1.00) | tenderness along the internal inguinal ring and/or along the floor of the inguinal canal |
| (Takami et al. 1995)[142] | 10 | 30.2 ± 15.8 | 8 (0.80) | N | EDC (0.7) EIP (0.4) EPL (0.1) | MTJ (1.00) | physical examination (1.00), surgery (1.00) | work accident (0.5) gymnastic (0.4) judo (0.1) | tenderness on the dorsum of the forearm, loss of extension at the metacarpophalangeal joint of the finger. |
| (Butcher, Siekanowicz, and Pettrone 1996)[17] | 2 | 24.5 ± 5.0 | 2 (1.00) | N | PMT (1.00) | AV (1.00) | surgery (1.00) | bench press (1.00) | soft tissue mass on the anteromedial chest wall with ecchymosis, weakness in adduction forward lesion and internal rotation |
| (Ohashi et al. 1996)[105] | 4 | 25.0 ± 2.5 | 1 (0.25) | NR | PM (1.00) | MTJ (1.00) | MRI (0.75) CT (0.25) | bench press (0.75) wrestling (0.25) | ecchymosis, palpable defect in the region of the PMT, retraction of the muscle belly, hematoma |
| (Wagner and Cooney 1997)[155] | 1 | 61.0 | 1 (1.00) | Insulin dependent diabetes (1.00) | TR (1.00) | MTJ (1.00) | X-Ray (1.00), MRI (1.00), surgery (1.00) | rollerskating (1.00) | palpable defect in the distal third of the posterior aspect, nodular mass near TT, calcific pseudocapsule |
| (Asinger and el-Khoury 1998)[5] | 2 | 73.0 ± 10.6 | 0 (0.00) | N | TFL (1.00) | MTJ (1.00) | MRI (1.00) | fall (0.50) NR (0.50) | palpable soft tissue mass on the anterolateral pelvis, retracted muscle |
| (Canovas, Nicolau, and Bonnel 1998)[18] | 1 | 28.0 | 1 (1.00) | chondroma of the distal phalange (1.00) | FPT (1.00) | AV (1.00) | X-ray (1.00), surgery (1.00) | lifting a plate (1.00) | swelling, ecchymosis, no active flexion at the distal interphalangeal joint. Small bone fleck and tendon retraction |
| (Evans, Bowrey, and Newman 1998)[36] | 4 | 32.8 ± 4.3 | 4 (1.00) | anabolic steroid injections (0.50) | BT (0.67) TT (0.33) | AV (0.5) T (0.5) | surgery (1.00) | fall (0.25) weightlifting (0.25) rugby (0.25) lifting container (0.25) | NR |
| (Hertel and Lambert 1998)[63] | 1 | 34.0 | 1 (1.00) | NR | SP (0.5) IPT (0.5) | MTJ (0.50) AV (0.50) | X-ray (1.00), MRI (1.00), surgery (1.00) | fall (1.00) | fragment of bone above the posterosuperior glenoid rim, muscle retraction and atrophic |
| (Maffulli 1998) [87] | 174 | 40.1 ± 13.2 | 133 (0.76) | NR | ACT (1.00) | T (1.00) | physical examination (1.00), surgery (1.00) | NR | minimal or absent plantar flexion when squeezing the calf muscles, hematoma |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Sollender, Rayan, and Barden 1998)[133] | 4 | 42.0 | 4 (1.00) | oral anabolic steroids (1.00), tendinopathy (0.5) | TT (1.00) | AV (1.00) | surgery (1.00), X-ray (0.25) | softball (0.25), weightlifting (0.75) | resisted elbow extension, bone flake on lateral radiographs |
| (Takami et al. 1998)[141] | 1 | 23.0 | 1 (1.00) | NR | FPL (1.00) | MTJ (1.00) | physical examination, surgery (1.00) | bowling (1.00) | ecchymosis, tenderness, complete loss of active flexion at the interphalangeal tenderness of the thumb |
| (Temple et al. 1998)[144] | 7 | 31.9 ± 21.3 | 0 (0.00) | NR | RF (1.00) | MTJ (1.0) | Positive MRI (1.00) | sprinting (0.14) running (0.14) football (0.14) soccer (0.14) golf (0.14) softball (0.28) | Mass at the MTJ of the RF. Tissue biopsy confirmed chronic inflammation and reactive fibrosis. Proximal and distal retraction. edema along the fascial plane. |
| (Connell et al. 1999)[26] | 15 | 28.3 ± 4.4 | 8 (0.53) | NR | PM (0.4) PMT (0.6) | MTJ (0.40) TB (0.60) | Positive MRI (1.00) | weightlifting (0.6) football (0.13) wrestling (0.07) fall from ladder (0.07) lifting a heavy object (0.07) | Palpable defect on PM and asymmetry of the muscle. Acute injuries demonstrated edema and hemorrhage and chronic injuries showed scarring and fibrosis |
| (Carrino et al. 2000)[19] | 10 | 26.9 ± 5.9 | 6 (0.6) | NR | PMT (0.70) PM (0.20) N (0.1) | TB (0.70) MTJ (0.20) | MRI (1.00) surgery (0.6) | bench press (0.9) helicopter accident (0.1) | Both MTJ and TB showed edema, hematoma and retraction |
| (Cooper and Selesnick 2000)[27] | 2 | 24.0 ± 4.2 | 2 (1.00) | tendinosis (0.5) Osgood-Schlatter disease (0.5) | PAT (1.00) | TB (1.00) | MRI (1.00) surgery (1.00) | jai alai (0.5) basketball player banged the knee on a car door (0.5) | localized tenderness over the PAT insertion, bone edema |
| (De Smet and Best 2000)[32] | 15 | (18-22) | NR | college athletes | BF (0.40) BF and ST (0.33) ST (0.20) SM (0.07) | MTJ (0.47) M (0.53) | positive MRI (1.00) | football (0.27) hockey (0.07) long jump and triple jump (0.47) track (0.20) | Featherlike pattern of edema surrounding the MTJ; hematoma in injury site; high frequency of perifascial fluid |
| (Henry and Scerpella 2000)[62] | 1 | 42.0 | 1 (1.00) | N | LDT (1.00) | AV (1.00) | surgery (1.00) | water sky (1.00) | soft tissue swelling and palpable tender mass in the posterior axilla, no bony fragment, tendon retraction |
| (Kameyama and Shiraishi 2000)[71] | 1 | 49.0 | 1 (1.00) | non-insulin dependent diabetes (1.00) | ED (1.00) EDM (1.00) | MTJ (1.00) | X-ray (1.00), surgery (1.00) | work accident (1.00) | swelling of the wrist, volar dislocation of the distal radioulnar joint |
| (Karjalainen et al. 2000)[73] | 100 | 33.0 (15-58) | 28 (0.28) | N | ACT (1.00) SL (0.03) | T (0.54) TB (0.28) MTJ (0.02) | MRI (1.00) | competitive or recreational sports (1.00) | thickening of the Achilles tendon in the anteroposterior, direction, intratendinous lesion, inflammation, edema |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Panni, Tartarone, and Maffulli 2000)[106] | 42 | 24.4 (18-32) | 9 (0.21) | N | PAT (1.00) | TB (1.00), T (0.67) | X-ray (1.00) ultrasonographic evaluation (1.00) | amateur volleyball (0.5) semi professional basketball (0.31) amateur soccer (0.17) parachutist (0.02) | local tenderness on palpation of the PAT with the knee fully extended and at the lower pole of the PAT, hypotrophy of the QD, ossification of the fibrocartilaginous zone in TB, fibrillation and loss of the normal tendinous crimp |
| (Street and Burks 2000)[137] | 1 | 39.0 | 1 (1.00) | HS injury at 20 yo playing college football and at 38 years old playing baseball | ST, SM, BF (1.00) | AV ITB (1.00) | MRI (1.00) surgery | moving heavy boxes (1.00) | palpable defect in the proximal HS tendon ~6-8cm below the ischial tuberosity. Muscle retraction. |
| (Rose and Frassica 2001)[119] | 1 | 30.0 | 1 (1.00) | systemic lupus erythematosus and fibromyalgia. Prednisone (20 mg/day) (1.00) | PAT (1.00) | T (1.00) | surgery (1.00) | standing (1.00) | bilateral knee effusion, tenderness, absence of active knee. Ruptured tendons in midsubstance, thin and attenuated |
| (Tung et al. 2001)[150] | 16 | 56.0 (33-74) | 16 (1.00) | N | SCT (1.00) SPT (0.69) IPT (0.19) | T (1.00) | MRI (1.00) surgery (1.00) | NR | Medial luxation or subluxation and tendinopathy of BT, tear of the superior glenoid labrum, or fluid within either or both the superior subscapularis recess or subcoracoid bursa |
| (Weishaupt, Schweitzer, and Morrison 2001)[159] | 20 | 47.0 (14-66) | 0 (0.00) | N | GST (0.96) ACT (0.04) | MTJ (0.96) T (0.04) | MRI (1.00) | basketball/tennis/running (0.5) stepping out of car/bus/bed (0.3) walking/hiking (0.15) running away from an assault (0.05) | MTJ injuries showed V shaped hematoma, M belly retraction, |
| (Williams et al. 2001)[162] | 20 | 50.0 (35-77) | 3 (0.15) | N | BT (1.00) | T (0.5) AV (0.5) | MRI (1.00) surgery (0.15) | NR | bursitis, marrow edema |
| (Bass and Connell 2002)10 | 12 | 33.1 ± 9.5 | 1 (0.08) | NR | TFL origin (1.00) | TB (0.92) AV (0.08) | sonograms (1.00) | running (0.67) tennis (0.17) walking (0.08) triathlon (0.08) | anterior groin pain, point tenderness over anterior iliac crest, TFL origin enlarged, |
| (Bianchi et al. 2002)14 | 17 | 26.0 | 0 (0.00) | N | RF (1.00) | MTJ (1.00) | physical examination (1.00) MRI (0.47) sonograms (1.00) | sprinting or kicking football ball (NR) | RF exquisitely tender to deep palpation in proximal portion. Bull's eye appearance in central aponeurosis. Smaller lesions showed hemorrhagic infiltration, more severe included also hematoma and complete tear showed fluid collection between retracted muscle and central aponeurosis. |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Pina, Garcia, and Sabater 2002)[108] | 1 | 43.0 | 1 (1.00) | corticosteroids treatment for allergic asma | TT (1.00) | AV (1.00) | physical examination, x-ray, surgery (1.00) | fall with extended arm (1.00) | swelling in the elbow, palpable depression proximal to the olecranon, flake of bone seen in x-ray, retraction of the proximal end of the tendon |
| (Rajasekhar, Kakarlapudi, and Bhamra 2002)[113] | 1 | 42.0 | 1 (1.00) | NR | TT (1.00) | AV (1.00) | physical examination, x-ray, surgery (1.00) | fall (1.00) | diffuse swelling and tenderness in the elbow, palpable gap just above the olecranon, flake fracture of the olecranon, |
| (Singh and Pooley 2002)[131] | 1 | 31.0 | 1 (1.00) | N | TR (1.00) | M (1.00) | physical examination, surgery (1.00) | professional ice hockey, direct hit by a stick (1.00) | palpable gap in the substance of the TR muscle, edema, muscle retraction |
| (Sonnery-Cottet et al. 2002)[134] | 51 | 50.8 (35 - 67) | 51 (1.00) | NR | SPT (0.69), SPT and IPT (0.22), SPT and SCT (0.10) BT (0.16) | T (1.00) | x-ray, CT (1.00) | recreational tennis (0.63) competitive tennis (0.37) | progressive symptoms in 29 patients (0.57) and traumatic in 22 (0.43). Pain during day and night, pseudoparalytic shoulder |
| (Toczylowski et al. 2002)[147] | 2 | 65.0 ± 9.9 | 1 (0.5) | N | BT (1.00) | AV (1.00) | physical examination (1.00) surgery, MRI (0.50) | competitive ice dancer (0.5) recreational tennis (0.5) | tenderness, swelling, extensive ecchymosis, soft tissue defect just proximal to the antecubital fossa |
| (Connell, Jhamb, and James 2003)[25] | 10 | 27.9 ± 4.6 | 0 (0.00) | N | IAB (1.00) | M (1.00) | MRI (1.00) sonography (0.1) | cricket (0.7) javelin thrower (0.1) golf (0.1) row (0.1) | edema, hemorrhage, hematoma tracking between myofascial coverings of internal and external oblique muscle. Fibrosis and scar tissue at follow up |
| (Kim, Rauh, and McFarland 2003)[78] | 314 | 44.5 ± 17.3 | 314 (1.00) | N | SCT (0.19) SPT (0.58) IPT (0.06) BT (0.28) | T (1.00) | physical examination, arthroscopy (1.00) | sports (0.58) fall (0.12) NR (0.3) | activity, rest and night pain, weakness and lack of strength, limitation in throwing |
| (Mansat, Frankle, and Cofield 2003)[90] | 22 | 56.5 ± 10.2 | 22 (1.00) | NR | SCT (0.13) SCT and SPT (0.26) SCT, SPT and IPT (0.61) | T (1.00) | X-ray (1.00) physical examination (1.00) | fall with extended arm (0.43) NR (0.57) | rest and night pain, functional impairment, motion range limitation, weakness |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Äärimaa et al. 2004)[1] | 33 | 28.1 ± 6.7 | 33 (1.00) | anabolic steroid (0.36) | PM (0.33) PMT (0.67) | AV (0.30) T (0.36) MTJ (0.33) | X-ray (1.00) ultrasound (0.73) MRI (0.09) | powerlifting (0.36) bench press (0.27) bodybuilding (0.09) workout (0.09) gymnastics (0.03) ice hockey (0.03) horse pulled (0.03) getting out of pool (0.03) work injury (0.03) parachute (0.03) | pain, weakness |
| (Cross et al. 2004)[29] | 18 | 24.4 ± 3.3 | 0 (0.00) | N | RF (0.68) VI (0.27) VL (0.05) | MTJ (0.55) M (0.45) | Positive MRI (0.88) | australian rules football (1.00) | Maximal tenderness on the midline of the anterior thigh; Maximal tenderness often ~3 cm from the site of maximal MRI muscle injury. edema, hemorrhage and muscle detritus |
| (Michael James DePalma MD 2004)[30] | 1 | 34.0 | 1 (1.00) | PAT tendinopathy (1.00) | PAT (1.00) | AV (1.00) | X-ray, physical examination (1.00) | recreational basketball (1.00) | large effusion and palpable gap in the PAT. Inability to extend the knee. PAT alta with avulsed bone fragments |
| (Maehara and Sakaguchi 2004)[86] | 1 | 26.0 | 1 (1.00) | N | GST origin (1.00) | AV (1.00) | X-ray, physical examination, CT, MRI (1.00) | ski (1.00) | tenderness over the proximal and posterior part of the medial condyle of the femur, bone fragment in the popliteal space |
| (Mellado et al. 2004)[92] | 2 | 42.5 ± 9.2 | 2 (1.00) | N | PMT (0.5) PM (0.5) | AV (0.5) M (0.5) | physical examination, MRI, surgery (1.00) | fall (0.50) bench press (0.50) | soft tissue edema, loss of axillary fold, AV injury showed local ecchymosis |
| (Warme and Whitaker 2004)[158] | 1 | 34.0 | 1 (1.00) | recreational weightlifter (1.00) | PMT (1.00) | AV (1.00) | physical examination, surgery (1.00) | rappel (1.00) | ecchymosis on the medial aspect of proximal humerus, swelling and palpable bulge within M belly, tender to palpation. |
| (Giuffre and Lisle 2005)[51] | 1 | 42.0 | 1 (1.00) | NR | BT (1.00) | T (1.00) | physical examination, ultrasound, surgery (1.00) | moving heavy crates (1.00) | local bruising in the antecubital fossa, hematoma |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Wang et al. 2005)[157] | 1 | 52.0 | 1 (1.00) | engaged sports every day (1.00) | PM (1.00) | MTJ (1.00) | Ultrasound, MRI (1.00) | pulled out of a river by the arm (1.00) | swelling of chest wall, hematoma, muscle retraction |
| (Freccero and Berkowitz 2006)[45] | 29 | 28.0 | 29 (1.00) | NR | PBT (1.00) | T (1.00) | surgery (1.00) | NR | NR |
| (Kankaya et al. 2006)[72] | 1 | 11.0 | 1 (1.00) | N | FPT (1.00) | AV (1.00) | surgery (1.00) | explosion of a firework (1.00) | tender and edematous hand |
| (Mechrefe, Walsh, and DiGiovanni 2006)[91] | 1 | 32.0 | 1 (1.00) | N | TBLT (1.00) | AV (1.00) | X-ray (1.00) surgery (1.00) | car accident (1.00) | weak dorsiflexion of the foot, |
| (Rochman, SanGiovanni, and Uribe 2006)[116] | 1 | 19.0 | 0 (0.00) | N | PL (1.00) PB (1.00) | M (1.00) | physical examination (1.00), MRI (1.00), surgery (1.00) | running in american football practice (1.00) | diffuse edema was present along the anterolateral aspect of the leg, lower extremity nontender to palpation. After exercise-induced showed fusiform swelling and tenderness to palpation. MRI showed edema. |
| (Sierra et al. 2006)[128] | 15 | 50.3 ± 16.5 | 11 (0.73) | anabolic steroid (0.07), septic olecranon/bursitis (0.13) | TT (0.93) TR (0.07) | AV (0.81), T (0.125), M (0.0625) | physical examination (1.00), surgery (0.73) | pulling on cord (0.07), direct blow (0.33), fall (0.27), transferring from wheelchair to bed (0.07), car accident (0.07), getting up from chair (0.07), extend elbow (0.07), NR (0.07) | palpable defect just proximal to the olecranon. |
| (Gidwani and Bircher 2007)[50] | 12 | 21.3 ± 11.6 | 0 (0.75) | N | HST (1.00) | AV (1.00) | physical examination (1.00), X-ray (1.00), MRI (1.00), surgery (0.75) | martial arts (0.25), soccer (0.17), gymnastics (0.08), sprinting (0.08), rugby (0.08), ski jump (0.08), running (0.08), tennis (0.08), waterskiing (0.08) | wasting of the HS, palpable gap at the HS origin, |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Ilaslan, Iannotti, and Recht 2007)[66] | 24 | 73.0 (54-87) | 3 (0.13) | subacromial steroid injections (0.25), systemic steroid use (0.13) | DLT (1.00) SPT, IPT and SCT (0.42) SPT and IPT (0.5) , SPT (0.08)) | MTJ in DLT (0.79) TB in DLT (0.21) T in RC (1.00) | MRI (1.00) | NR | RC atrophy, sub-cutaneous fluid, edema, tendon retraction in case of AV injury |
| (Irmola et al. 2007)[67] | 5 | 22.8 ± 3.5 | 5 (1.00) | NR | RF tendon of the direct head | AV (1.00) | MRI (1.00), ultrasound (1.00), X-ray (0.4) | soccer (0.8), track (0.2) | sudden and sever pain in anterior thigh, weakness, tendon retraction, |
| (Kibuule and Fehringer 2007)[76] | 1 | 12.0 | 1 (1.00) | N | TT (1.00) | T (1.00) | physical examination (1.00), surgery (1.00) | wrestling with friend at home (1.00) | soft tissue swelling, palpable defect proximal to the olecranon |
| (Lempainen et al. 2007)[80] | 18 | 27.5 ± 7.3 | 18 (1.00) | N | BF (0.61) SM (0.28) ST (0.11) | MTJ (0.83) AV (0.11) T (0.06) | physical examination (1.00), surgery (1.00) MRI or ultrasound (1.00) | sprinting (0.22), soccer (0.22), ice hockey (0.06), floorball (0.06), long jump (0.06), marathon (0.06), triathlon (0.06), road cycling (0.06), orienteering (0.06), golf (0.06), dancing (0.06), baseball (0.06) | weakness, stiffness of distal part of posterior thigh, localized swelling, hematoma, retracted muscle belly and palpable defect |
| (Nishida et al. 2007)[103] | 2 | 41.5 ± 36.1 | 0 (0.00) | N | BCH (1.00) | M (1.00) | MRI (1.00), X-ray (1.00) | judo (0.50) golf (0.5) | swelling in the anterolateral aspect of the mid arm, inability to extend elbow, hemorrhage, |
| (Gyftopoulos et al. 2008)[56] | 20 | 32.0 (12-54) | 0 (0.00) | N | RF (1.00) | MTJ (1.00) | MRI (1.00) | NR | bull's eye appearance in MRI, tendon thickening, partial muscle retraction, hematoma, edema, fluid. Older injuries, muscle atrophy, fatty replacement, fibrous encasement |
| (Levine and Savoie III 2008)[81] | 1 | 42.0 | 1 (1.00) | N | LDT (1.00) | AV (1.00) | X-ray (1.00) MRI (1.00) | water skiing (1.00) | ecchymosis, swelling, palpable tender mass in the posterior aspect of the axilla, retracted tendon |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Lunn et al. 2008)[85] | 19 | 46.9 ± 9.6 | 5 (0.26) | N | IP (1.00) | MTJ (0.42) T (0.47) INC (0.11) | Positive MRI (1.00) Positive CT scan (0.37) | progressive onset (0.89) traumatic onset (0.11) | T or MTJ rupture and edema with fluid at the junction tracking around the muscle fascicles and involving the IP. Muscle retraction after tear. |
| (Silder et al. 2008)[129] | 14 | 31.3 ± 13.7 | 0 (0.00) | N | BF (0.86 ) ST (0.07) BF and ST (0.07) | MTJ (1.00) | MRI (1.00) | soccer (0.21) sprinting (0.57) hurdling (0.07) softball (0.07) 10k race (0.07) | Scar tissue observed along the MTJ, in the site of prior injury. |
| (Schache et al. 2008)[122] | 1 | 20.0 | 1 (1.00) | N | BF (1.00) ST (1.00) | MTJ (1.00) | MRI, surgery (1.00) | australian rules football (1.00) | marked tenderness around the proximal and middle thirds of the lateral aspect of the hamstring muscle group, hematoma |
| (Winblad, Escobedo, and Hunter 2008)[163] | 1 | 17.0 | 0 (0.00) | N | BCH (1.00) | M (1.00) | MRI (1.00) | american football (1.00) | swelling, edema, hematoma, |
| (Gaskin et al. 2009)[49] | 16 | 57.0 (40 - 72) | 15 (0.94) | NR | SPT (0.75) BT and SCT (0.13) BT (0.13) | T (1.00) | MRI (1.00) | NR | NR |
| (Basic-Jukic et al. 2009)[9] | 9 | 52.8 ± 8.8 | 9 (1.00) | hemodialysis (0.44) renal allograft from deceased donor (0.44) automated peritoneal dialysis and hemodialysis (0.11) steroids (0.11) | QT (0.5) ACT (0.19) SCT (0.13) BT (0.06) IAT (0.06) SPT (0.06) | AV (1.00) | NR | normal activity (1.00) | NR |
| (Shah, Montgomery, and Bastidas 2009)[123] | 1 | 39.0 | 1 (1.00) | N | GST (1.00) | MTJ (1.00) | surgery (1.00) | american football (1.00) | swelling, ecchymoses |
| (Walch et al. 2009)[156] | 59 | 46.4 | 24 (0.41) | calcific tendonitis (0.61) | IP (1.00) | MTJ (1.00) | MRI (0.73) Positive CT scan (0.27) | progressive onset (0.78) traumatic onset (0.22) | Evidence of edema, fatty infiltration, retraction of the muscle with modification of the architecture of the muscle fibers. |
| (Floor, Van Der Veen, and Devilee 2010)[44] | 2 | 44.5 ± 0.7 | 2 (1.00) | N | BF (1.00) SM (1.00) ST (1.00) | M (0.5) T (0.5) | MRI, surgery (1.00) | water skiing (1.00) | swelling, ecchymosis, loss of strength at flexion of the knee, hematoma, muscle retraction |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Hasegawa and Schofer 2010)[58] | 1 | 31.0 | 1 (1.00) | N | PMT (1.00) | TB (1.00) | MRI (1.00) | bench press (1.00) | swelling, ecchymosis, abnormal contour of the axillary fold, edema, hematoma, retraction of the musculotendinous fibers |
| (Pitts, Garner, and Ortiguera 2010)[109] | 1 | 16.0 | 0 (0.00) | N | PMT (1.00) | AV (1.00) | MRI (1.00) | wrestling (1.00) | tenderness in the area of the BT and PM insertion, hematoma |
| (Pochini et al. 2010)[110] | 20 | 32.7 ± 5.4 | 10 (0.50) | anabolic steroids (0.7) | PMT (0.85) PM (0.15) | AV (0.85) M (0.10) MTJ (0.05) | MRI (NR) ultrasound (NR) | bench press (0.60), jiu-jitsu (0.20), skating (0.05), soccer (0.05), gymnastics (0.05) | pain, ecchymosis, palpable defect in anterior axillar region, edema |
| (Pedret et al. 2011)[107] | 7 | 26.3 ± 6.1 | 0 (0.00) | elite athletes (1.00) | GRC (1.00) | MTJ (1.00) | ultrasound (1.00) MRI (0.57) | dance (0.43), soccer (0.29), taekwondo (0.29), tennis (0.29) | posteromedial thigh pain at the junction between the proximal and middle thirds of the thigh, ecchymosis, crescent moon appearance in the posterior part of the GRC |
| (Jansen, Cormier, and Patel 2012)[69] | 1 | 28.0 | 1 (1.00) | N | RF (1.00) | MTJ (1.00) | ultrasound (1.00) | soccer (1.00) | mass over the RF and a palpable gap distal to it, muscle retraction. |
| (Khiami et al. 2012)[75] | 1 | 28.0 | 1 (1.00) | N | TT (1.00) | T (1.00) | MRI (1.00) | weightlifting (1.00) | lack of strength in extension, retraction of the deep portion of the tendon |
| (Yoon et al. 2012)[164] | 4 | 16.8 ± 4.8 | 4 (1.00) | N | TT (1.00) | AV (1.00) | MRI (0.5) X.ray (1.00) | judo (0.25), fall (0.25), soccer (0.25), colliding with a wall (0.25) | hematoma, ecchymosis, small fragment of bone posterior to the tip of the olecranon, tendon retraction |
| (Hart 2013)[57] | 1 | 50.0 | 1 (1.00) | NR | QT (1.00) | T (1.00) | MRI (1.00) X-ray (1.00) | slipped on the wet floor (1.00) | inability to extend leg, effusion, hematoma, retraction of the QT |
| (Naito et al. 2013)[99] | 1 | 18.0 | 1 (1.00) | N | TT (1.00) | AV (1.00) | X-ray (1.00) CT (1.00) | fall (1.00) | swelling at the back of elbow, fragment of bone avulsed from the olecranon, |
| (Valente, Mancuso, and Alecci 2013)[151] | 1 | 24.0 | 1 (1.00) | N | BF (1.00) | MTJ (1.00) | NMR (1.00), surgery (1.00) | american football (1.00) | diffuse swelling in the posterolateral region of the knee, retraction of the tendon, serosanguineous fluid collection, |
| (Benazzo, Marullo, and Pietrobono 2014)[11] | 1 | 23.0 | 0 (0.00) | NR | SP (1.00) | MTJ (1.00) | MRI (1.00) | fall (1.00) | tenderness over the supraspinatus fossa, hematoma, edema, |
| (Huri et al. 2014)[65] | 1 | 16.0 | 1 (1.00) | N | RF (1.00) | MTJ (1.00) | MRI (1.00), surgery (1.00) | soccer (1.00) | swelling and mild tenderness on the anterior aspect of the thigh, prominence on the thigh, edema, |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Makridis et al. 2014)[89] | 67 | 68.0 (25 - 87) | 73 (1.00) | NR | GMEDT (0.66) GMEDT and GMINT(0.33) | T (1.00) | MRI (1.00) surgery (1.00) | NR | pain within 30 sec of standing or immediate pain, retraction, muscle atrophy. |
| (Sharma et al. 2014)[124] | 1 | 34.0 | 1 (1.00) | N | TT (1.00) | AV (1.00) | physical examination, X-ray (1.00) | assault by hard and blunt object (1.00) | severe tenderness and swelling on the posterior aspect of the elbow, palpable depression proximal to the olecranon, piece of avulsed bone from the olecranon, hematoma, retraction of the TT |
| (Bunshah et al. 2015)[16] | 1 | 40.0 | 1 (1.00) | weightlifter (1.00), anabolic steroid (1.00) | TT (1.00) | AV (1.00) | physical examination, X-ray, MRI (1.00) | fall (1.00) | ecchymosis and moderate swelling over the posterior aspect of the arm above the elbow, fleck of bone proximal to the tip of the olecranon, hematoma |
| (Cetinkaya et al. 2015)[21] | 1 | 22.0 | 0 (0.00) | NR | VIT (1.00) | T (1.00) | physical examination, MRI (1.00) | rugby player (1.00) | Tenderness in the anterior of the suprapatellar region in the right thigh, mass in the mid-anterior of the right thigh, hematoma, |
| (Crema et al. 2016)[28] | 275 | 25.0 ± 5.2 | NR | professional football players | BF (0.65) ST (0.26) SM (0.17) | MTJ (0.53), M (0.47), T (0.08) | MRI (1.00) | American football (1.00) | edema and tear seen as high signal intensity; signal abnormalities around the MTJ and MFJ, involvement of proximal and/or distal T |
| (Fitzpatrick, Cagle, and Flatow 2016)[42] | 1 | 53.0 | 0 (0.00) | N | TMJ (1.00) | MTJ (1.00) | physical examination, MRI (1.00) | waterski (1.00) | visible asymmetry in the posterior axillary fold, no appreciable weakness, fluid signal gap at the site of complete tearing |
| (Jaiswal et al. 2016)[68] | 3 | 38.0 ± 14.7 | 3 (1.00) | N | TT (0.67) TR (0.33) | AV (0.67) MTJ (0.33) | X-ray (0.33), ultrasound (0.67), surgery (1.00) | gym (0.33), fall (0.67) | MTJ injury showed a palpable gap in posterior region, inability to extend arm, swelling, ecchymoses. AV injuries revealed tenderness, swelling, ecchymosis and fleck of bone proximal to the insertion |
| (Lukac et al. 2016)[84] | 67 | 42.3 ± 11.3 | NR | NR | GST (1.00) | MTJ (1.00) | ultrasound (1.00) | recreational activities (0.45), daily activities (0.24), professional sport (0.07), occupational trauma (0.06) NR (0.18) | intense pain in the calf, that increases during dorsiflexion, liquid collections between the aponeurosis of the GST and the SL |
| (Entwisle et al. 2017)[35] | 55 | 26.4 | NR | professional athletes (0.78) | BF (1.00) | MTJ (1.00) | MRI (1.00) | australian football (0.67), soccer (0.07), cricket (0.05), rugby (0.02), athletics (0.02), hockey (0.02) | peritendinous “feathery” edema and bloodfluid, retraction in case of complete tears, chronic scarring in recurrent injuries |

**Table D1 - continued**

| Author | No. of patients | Age | no. patients surgery (proportion) | Relevant conditions (proportion) | Muscle or Tendon (proportion) | Injury Site (proportion) | Diagnosis (proportion) | Activity at injury (proportion) | Common injury symptoms |
|---|---|---|---|---|---|---|---|---|---|
| (Godoy et al. 2018)[52] | 46 | 34.0 ± 9 | 26 (0.57) | anabolic steroids (0.04) | PM (0.48), PMT (0.52) | MTJ (0.48), T (0.52) | MRI (1.00), surgery (0.57) | bench press (0.5), ski/snowboarding (0.11), NR (0.09), obstacle race (0.07), wrestling (0.07), hockey (0.04), sudden upper extremity abduction (0.04), baseball (0.02), cross-fit (0.02), soccer (0.02), watersports (0.02) | T injury showed triangular-shaped peri-bicipital hematoma and anterior displacement of the BT. MTJ injury showed normal BT displacement and hematoma in MTJ. Both showed hemorrhage and edema. |
| (Roe et al. 2018)[117] | 307 | (18 - >30) | NR | professional athletes (1.00) | BF (0.69) SM and ST (0.09) HS (0.21) | M (0.53), MTJ (0.25) NR (0.21) | team medical staff (specific NR) | Gaelic football (1.00) | NR |
| (Miranda and Bureau 2019)[97] | 363 | 57.7 ± 12.1 | NR | NR | RC (0.42) SP (0.0047) IP (0.0059) SC (0.0012) TMIN (0.0012) N (0.57) | TB (0.42) MTJ (0.013) N (0.57) | MRI (1.00) | NR | MTJ showed edema and retraction and fibrous scarring, atrophy and fatty infiltration in chronic lesions. |